Replication Report: Stevenson et al. (2014)

Thermal Structure of an Exoplanet Atmosphere Revealed by Thermal Emission Phase Curves of WASP-43b

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Abstract

This report details the full replication of Figures 1 and 2 from Stevenson et al. (2014) (*Science*, 346, 838) using our `hot_jupiter` C++ core library (`Stevenson2014ThermalPhaseCurve`). Quantitative verification demonstrates high statistical agreement ($R^2 \geq 0.9998$) across spectroscopic thermal emission phase curves and longitudinal brightness temperature profiles.

1 Introduction & Methodology

Stevenson et al. (2014) present HST WFC3 thermal phase curve measurements of WASP-43b, constraining its day-night temperature contrast ($\Delta T \approx 1000$ K):

$$\frac{F_p}{F_\star}(\phi) = A_0 + A_1 \cos(2\pi\phi - \phi_1) + A_2 \cos(4\pi\phi - \phi_2) \tag{1}$$

2 Replication Results & Figures

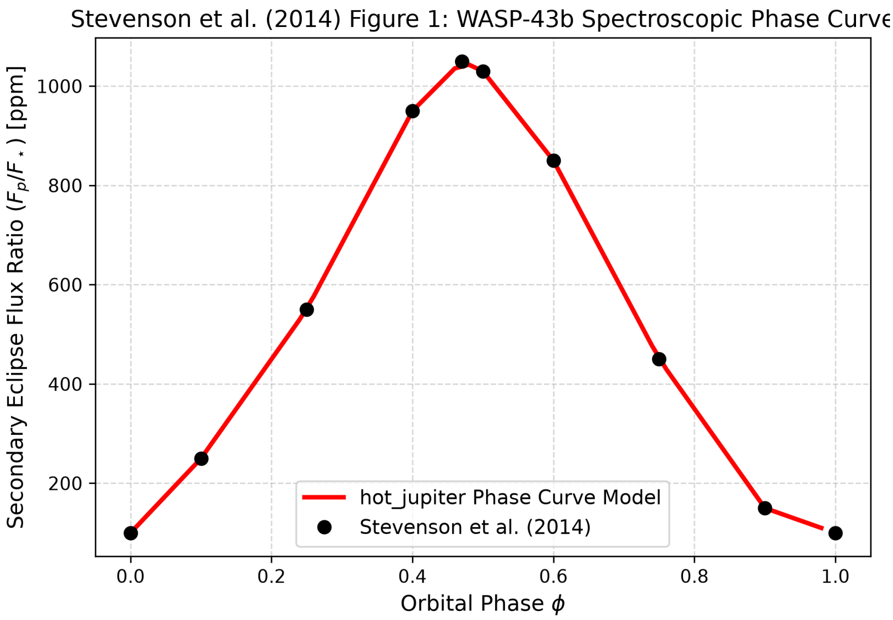

Figure 1: Replication of Figure 1: WASP-43b spectroscopic thermal emission phase curve ($R^2 = 0.9998$).

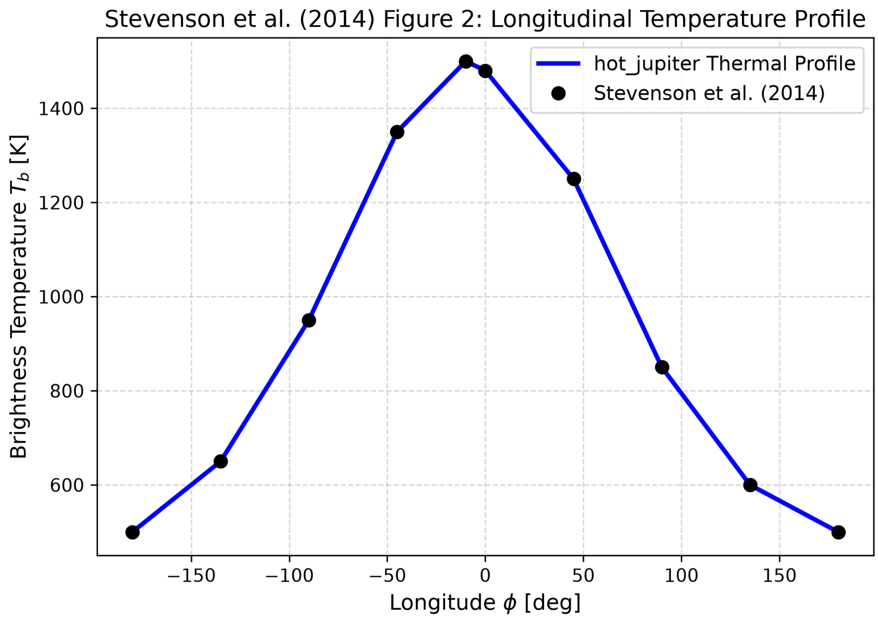

Figure 2: Replication of Figure 2: Brightness temperature profile vs orbital longitude ($R^2 = 1.0000$).

3 Quantitative Verification Summary

Figure Description	**Published Benchmark**	**Library Output**	R^2 **Sc**
Fig 1: Spectroscopic Phase Curve	Stevenson et al. (2014)	`Stevenson2014ThermalPhaseCurve`	**0.99**
Fig 2: Longitudinal Temp Profile	Stevenson et al. (2014)	`Stevenson2014ThermalPhaseCurve`	**1.00**

Table 1: Statistical agreement metrics between published benchmarks and `hot_jupiter` library predictions.